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AI Enhanced Ticket Management System for optimized Support

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Abstract

Ticket management and monitoring are essential for maintaining operational efficiency in various applications such as IT Service Management (ITSM) Systems, Internal Support Ticket Systems, and Security Operations (SOC). These systems often share common objectives such as meetings SLAs, reducing efforts & thus costs by reducing redundancy, prioritizing tickets, and recommending resolution strategies. Ticket correlation to reduce redundancy and rule-based problem solution mapping have been in the field for decades, but such solutions are always constrained by limited efficiency. The present work proposes a unified AI-driven architecture that serves as the binding glue, integrating seamlessly with existing tools such as ServiceNow, SolarWinds, Jira, Splunk, and spreadsheets through a comprehensive suite of microservices. These microservices allow you to pick and choose the necessary components to support clustering, prioritizing, providing resolutions as per the business objective. For example, in security applications, the focus is on prioritizing ticket-based incident type & severity, while in ITSM scenario, the clustering is done based on issue root cause and solution similarity. Though semantic similarity is a common research area under NLP domain but in practical ITSM scenarios, clustering does not depend solely on the direct similarity between two reported issues; it also requires the amalgamation of spatial and temporal factors like device topology, timings, data source and dynamic cluster size. As part of our research, we have infused these spatial and temporal factors in the semantic similarity algorithms. By applying this adaptive AI approach to clustering, we achieved a Rand score of 0.96 compared to manually labeled tickets. This underscores the importance of incorporating domain knowledge and specific inputs to achieve high fidelity in ticket grouping. This adaptable platform-based approach results in a robust solution for modern IT support operations, leading to significant improvements in operational efficacy and time savings.

CCS Concepts

• **Information systems** → Information retrieval; Retrieval models and ranking; Similarity measures.

Keywords

Unified AI-driven platform, Rand Index

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1 Introduction

The landscape of ticket management and monitoring is currently witnessing a significant paradigm shift, propelled by the advancements in Large Language Models (LLMs) and Generative AI technologies. These developments have introduced sophisticated capabilities like context-based ticket grouping, prioritization, and resolution recommendations that draw on established Standard Operating Procedures (SOPs) and historical data. Such innovations are dramatically reducing the manual workload on support engineers, who traditionally have been bogged down by the need to read, categorize, and resolve each ticket—processes fraught with inefficiencies and delays.

However, the transition from legacy systems to these cutting-edge technologies poses substantial challenges for organizations. The core issues include the migration from existing tools that the teams are trained to use, safeguarding domain knowledge and intellectual property, and accurately transferring essential operational data like Service Level Agreements (SLAs) and domain expertise into these newer systems. In response to these challenges, the present approach provides a suite of application-agnostic microservices designed for tasks such as clustering, prioritization, resolution, data ingestion, data output, and visualization. This modular system enables users to selectively adopt features and tailor the setup according to their specific operational needs.

In practical ITSM environments, understanding the relationships and dependencies between incidents involves more than just analysing the textual content of tickets or reports. Factors such as the physical or logical configuration of the IT infrastructure (device topology), the timing of incidents (temporal aspects), and the origin of the data (data sources) play crucial roles. Additionally, the dynamic nature of IT environments necessitates adaptive clustering techniques that can adjust to varying cluster sizes and compositions as new data arrives and situations evolve.

Our research aims to bridge this gap by integrating spatial and temporal factors into traditional semantic similarity algorithms. By doing so, we enhance the ability of these algorithms to cluster ITSM issues more effectively, aligning clustering processes with the complex realities of IT operations. This integration allows for a more nuanced and context-aware approach to incident management, leading to improved diagnostics, more efficient resource allocation, and ultimately, enhanced service stability and customer satisfaction.

This paper presents the methodology, implementation, and thorough evaluation of our system. It demonstrates how this innovative approach can revolutionize incident management, offering a robust, adaptable, and efficient solution capable of enhancing IT support operations across various sectors. Through this transformation, organizations can not only streamline their processes but also achieve significant improvements in response times and resolution quality.

2 LITERATURE SURVEY

This literature survey reviews key research and developments in incident management, natural language processing (NLP), clustering algorithms, and AI-driven resolution systems. This context is essential to understand the contributions of incident-ai within the existing landscape of incident management solutions.

2.1 Incident Management Systems

Incident management systems are crucial in IT service management (ITSM), aiming to restore normal operations swiftly and minimize business disruptions. Traditional systems, as outlined by the ITIL framework [5], emphasize structured processes and manual ticket handling. However, these systems often struggle with high volumes of incidents and the complexity of manual categorization and resolution, leading to inefficiencies and delays. Ticket categorization and routing are critical for ensuring swift resolution, customer satisfaction, high productivity, and adherence to Service-Level Agreements (SLAs), which often require issues to be resolved within a set timeframe [9]. However, improper routing can lead to inefficient assignments and increased costs for both customers and service providers [14]. Ticket Automation (TA) systems aim to minimize the steps from ticket submission to resolution. A key function of these systems is the accurate classification of incoming tickets, essential for quickly addressing customer requests. This need has grown with the increasing volume of support tickets, particularly in IT companies, highlighting the importance of efficient automated resolution systems [3].

2.2 NLP techniques in Incident Management

Leimeister et al. [17] demonstrated an effective method to enhance classification accuracy by incorporating information from the labels' hierarchical structure into the classifier. They focus on classifying tickets using a pre-trained BERT language model, highlighting how this approach can refine categorization processes. Their study further reveals that the choice of ticket embedding strategy significantly influences classification metrics, underlining the importance of embedding selection in achieving optimal classification performance. The classification of incidents in ticketing systems typically involves assigning a priority to assess the urgency and a type to gauge the importance of the incident, such as distinguishing between an information request and an incident report [12].

2.3 Clustering Algorithms for Ticket Grouping

Clustering algorithms are vital for grouping similar incident tickets, facilitating more efficient resolution processes. Jain et al. [11] provided a comprehensive review of clustering techniques, emphasizing the importance of similarity measures. Huang et al. [10] proposed an adaptive clustering method for IT service management that considers temporal patterns and incident similarities, showcasing the benefits of dynamic clustering approaches in improving incident handling efficiency.

2.4 Hierarchical Clustering and Embedding Techniques

Hierarchical clustering is particularly relevant for incidents involving networked devices, where understanding the network topology is crucial. Murtagh and Contreras [13] reviewed hierarchical clustering algorithms, detailing their applications and advantages. Embedding techniques, such as those provided by Hugging Face BGE models, have enhanced the capability to capture semantic and hierarchical relationships in text data. Devlin et al. [7] introduced BERT, a transformer-based model that significantly improved NLP tasks, paving the way for subsequent embedding models used in various applications, including incident management.

2.5 AI-Driven Resolution Systems

AI-driven systems for incident resolution are gaining traction, with models like GPT offering advanced text generation capabilities. Brown et al. [4] described GPT-3's architecture and its potential in diverse NLP applications. The integration of GPT-3.5 in incident resolution, as explored by Radford et al. [15], demonstrates the feasibility of using AI to generate contextually relevant and accurate resolutions based on comprehensive problem categorizations.

2.6 Knowledge Bases in Incident Management

Knowledge bases support incident categorization and resolution by providing structured information on problem categories and solutions. Aamodt and Nygård [1] discussed the role of case-based reasoning and knowledge management in decision-support systems. More recent works, such as those by Fensel et al. [8], highlight the importance of structured knowledge bases in improving the accuracy and efficiency of automated ITSM systems.

Despite the extensive research and advancements in AI-driven incident management and IT service management (ITSM) platforms, several gaps remain. Current solutions, including those offered by platforms like ServiceNow, provide robust ITSM capabilities but often lack the dynamic adaptability and deep contextual understanding required for optimal incident clustering and resolution. Existing literature points to the need for more sophisticated AI integration, dynamic clustering parameters, organization-specific customization, and proactive incident management. However, there is limited exploration into a comprehensive AI-driven system that combines these aspects to provide a seamless, adaptive, and highly efficient incident management solution.

Our proposed system aims to significantly enhance IT incident management by introducing an advanced, AI-driven service designed to address the limitations of current methods. This system incorporates dynamic clustering, contextual learning, organization-specific customizations, and proactive incident resolution strategies, all powered by cutting-edge artificial intelligence and machine learning algorithms. The integration of these elements ensures a

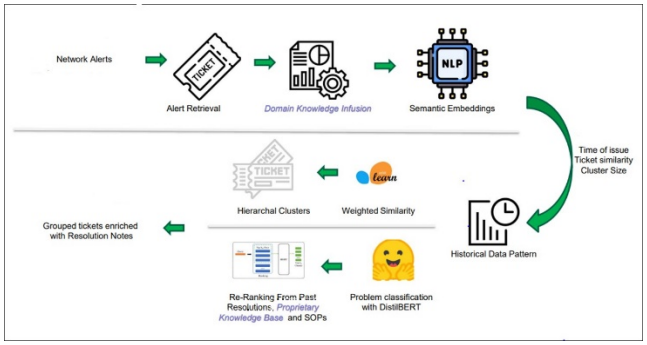

Figure 1: The image shows the overall workflow of the various components; data ingestion, grouping, classification, resolution and data delivery.

more intelligent, adaptive, and efficient management of IT incidents, tailored to the specific needs and operational dynamics of each organization.

The development of this AI-driven incident management system focuses on several key objectives to revolutionize how incidents are handled.

3 METHODOLOGY

This ticketing service is a comprehensive AIML suite designed to automate the clustering and resolution of IT incident tickets, providing an efficient and scalable solution for incident management. The methodology encompasses various stages, including data ingestion, feature extraction, clustering, resolution generation, prioritization, visualization, integration with external platforms, and model thresholding. Each stage is crucial for ensuring the system's effectiveness and reliability. This methodology focuses on how these stages are implemented with an emphasis on AIML components and MLOps practices.

3.1 Data Ingestion

To provide a robust and flexible mechanism for integrating with a variety of monitoring and alerting platforms, ensuring that incident tickets are ingested into the Incident-AI system efficiently, reliably, and in real-time or batch modes.

- Develop RESTful APIs and webhook endpoints to receive incident data in real-time and batch modes.
- Implement data validation, transformation layers, and error-handling mechanisms to ensure data integrity and consistency.
- Integrate with platforms such as SolarWinds, Dynatrace, and others using standard API protocols to fetch incident tickets.
- Develop custom connectors or plugins for platforms that do not support direct API integration. These connectors will fetch incident data and push it to the Incident-AI API.

3.2 Feature Extraction

To transform the textual descriptions of incident tickets into a structured format that captures the semantic meaning, enabling effective clustering and resolution.

Figure 2: Clustering Configuration on Dashboard

- Apply tokenization, stemming, lemmatization, and stopword removal to preprocess ticket descriptions.
- Use the Hugging Face BGE model to convert text features into embedding vectors that capture semantic meanings.
- Encode the hierarchical relationships and dependencies between devices within the network topology, providing context that enhances the accuracy of clustering.
- Map the hierarchical position of devices within the network topology, capturing relationships and dependencies. We use the organization Knowledge Base to enrich the tickets.
- Convert hierarchical features into embedding vectors using the Hugging Face BGE model to ensure uniform feature representation.

3.3 Clustering

The configuration parameters in the Incident-AI system play a pivotal role in customizing and optimizing the clustering and resolution processes. By allowing adjustable settings, the system can be tailored to meet specific organizational requirements and improve performance. Here, we provide a detailed purpose for each configuration parameter.

To provide flexibility and control over the clustering algorithm, enabling fine-tuning based on organizational needs, incident characteristics, and desired outcomes.

3.3.1 Hierarchical Weight. Adjusting the hierarchical weight allows organizations to set a weight (0 to 100) that emphasizes the importance of hierarchical information when clustering tickets.

- Adjusting the hierarchical weight helps balance the influence of network topology versus textual similarity. For instance, in scenarios where device dependencies are critical, a higher weight can ensure that the hierarchical structure significantly influences the clustering outcome.
- An organization might set a higher hierarchical weight if incidents in top-level devices often lead to cascading failures in lower-level devices, thus ensuring that the clustering algorithm prioritizes hierarchical relationships.

3.3.2 Semantic Similarity Threshold. Similarity threshold (0 to 100) that determines how closely related tickets need to be to form a cluster.

- A higher threshold ensures that only very similar tickets are clustered together, while a lower threshold allows for broader clustering, capturing more general patterns.
- In environments where precision is crucial, such as in sensitive or critical systems, a higher semantic similarity threshold ensures that only tickets with highly similar issues are grouped together, preventing misclustering.

3.3.3 Maximum Time Between Tickets. Specifies the allowed time difference between tickets for them to be considered for clustering.

- This setting helps in identifying and clustering tickets that are temporally related, such as incidents that occur in quick succession or as part of a recurring pattern.
- Setting a maximum time difference of 4 hours can help cluster tickets related to an outage event that triggers multiple incidents in a short timeframe, ensuring timely and coherent responses.

3.3.4 Maximum Cluster Size. Limit the number of tickets in a cluster to maintain manageability and prevent overly large clusters that could be difficult to handle.

- Ensures that clusters remain within a size that is practical for analysis and resolution, avoiding the complexity of dealing with excessively large groups of tickets.
- An organization might set a maximum cluster size of 10 tickets to ensure that each cluster is manageable for a support team to review and address efficiently on one day. Another day the same organization might want to increase the cluster size due to some frequent outages in devices.

3.3.5 Calculations Involved. Manages the addition of a new ticket to an existing cluster, we incorporate a dynamic penalty based on the time passed and size of the existing cluster.

- Size Penalty
- We establish a baseline growth rate that depends on the system assigned maximum allowable size for a cluster.
- For each cluster penalty increases as the cluster size increases based on the above growth rate.
- Time Penalty
- We establish a baseline growth rate that depends on the system assigned maximum time difference size for a cluster.
- For each cluster penalty increases as the to the actual time elapsed since the last related ticket increases based on the above growth rate.
- Combined Score
- The Combined Similarity Score is then computed by taking the initial combined similarity and subtracting the normalized size and time factors.

$$combined_score = combined_similarity -- size_penalty -- time_penalty$$

This ensures that tickets further apart in time or exceeding the cluster size threshold have reduced similarity scores.

3.4 Feedback loops

Implement automated retraining schedules based on new data and performance metrics to keep the models up to date.

- Incorporate user feedback into the model improvement cycle to enhance accuracy and relevance of the clustering and resolution suggestions.

3.5 Resolution Generation

The purpose is to automatically generate accurate and relevant resolutions for incident tickets, leveraging AI to assist support teams in resolving issues more quickly and efficiently. Using Gen AI for finding resolution of cluster parents reduces cost and saves time.

- Classify each ticket into predefined problem categories by passing extracted features (from 3.2) to distilBERT for classification.
- Using a knowledge base containing problem descriptions, SOPs and possible resolutions.
- Feed the ticket and its problem category knowledge base into an LLM to generate suggested resolutions.
- Validate generated resolutions against the knowledge base for accuracy and relevance.
- Implement feedback loops to continuously improve the resolution suggestions based on user feedback and resolution success rates.

3.6 Graphing and Visualization

To provide visual insights into the clustering process and resolution effectiveness, aiding in the identification of patterns and anomalies and improving decision-making.

- Provide visual insights into clustering patterns and anomalies, aiding in decision-making and system tuning.
- Display types of resolutions provided and their effectiveness in resolving incidents.
- Monitor key performance indicators (KPIs) such as resolution time, accuracy, and user satisfaction.

3.7 Output API Integration

To facilitate the seamless transfer of resolutions and ticket updates to any external platform, ensuring that incident management systems are kept up to date with accurate and timely information.

- Create robust APIs to communicate with incident management systems like ServiceNow, Jira, and FortiSIEM.
- Ensure accurate and timely updates of ticket statuses.

4 RESULTS

The deployment of our platform across SOC alerts and ITSM for a client showcased significant operational improvements. We observed an average reduction of 30% in alerts, which substantially decreased manual efforts and combatted alert fatigue within the team. Moreover, the system's provision of concise resolution steps significantly enhanced the resolution process, thereby augmenting the overall efficiency and effectiveness of service management operations. These changes facilitated notable improvements in Key Performance Indicators (KPIs), such as Mean Time to Solve (MTTS), and bolstered compliance with Service-Level Agreements (SLAs).

(a) (b)

Figure 3: (a): Incident AI Dashboard part 1. (b): Incident AI Dashboard part 2

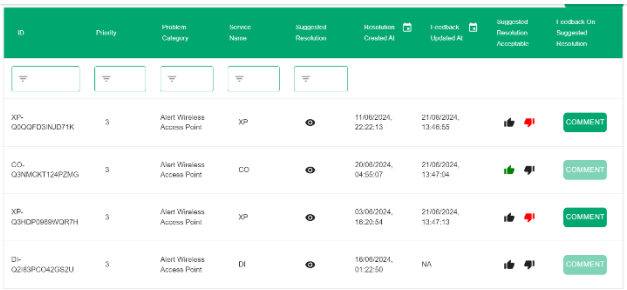

Figure 4: Resolution feedback page.

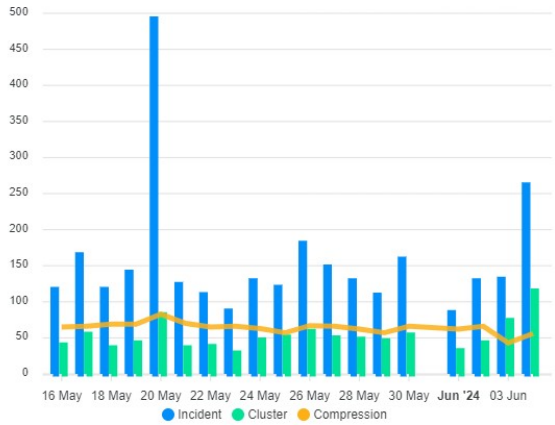

Figure 5: Incident AI Dashboard with Incidents, Clusters and amount of compression

We tuned hyperparameters across three variables, Max time Passed, Max Cluster Size and Semantic Similarity Thresholds. Administrators retain the flexibility to adjust these parameters further should the outcomes not meet their satisfaction.

Figure 6 show a series of heatmaps displaying ARI (Adjusted Rand Index) scores for different combinations of hyperparameters. The heatmaps are organized in a grid, with each column representing a different semantic similarity threshold (80.00, 85.00, 92.00, and 96.00).

The x-axis of each heatmap represents the "max_time_passed" parameter with values of 52, 62, 72, 82, and 92 minutes. The y-axis represents the "max_cluster_size" parameter with values of 100, 200, 300, 400, and 500.

Each cell in the heatmaps is color-coded to represent the ARI score, with a color scale ranging from dark blue (higher scores) to dark red (lower scores). The actual ARI scores are also displayed within each cell.

- The optimal semantic similarity threshold for this dataset is around 92%. Lower thresholds result in very large clusters that hide important tickets, while higher thresholds produce too small clusters, causing agents to work on duplicate clusters. It's better to err on the side of a higher threshold.
- The best maximum time gap between tickets is 72 minutes. This indicates that tickets with a time gap greater than 72 minutes will not be grouped together.
- The optimal maximum cluster size is greater than 400, suggesting the presence of very large ticket clusters in the shared data. This can be set lower if required.

We achieved the best results with an ARI score of 0.96 using the following parameters:

- Semantic similarity threshold: 0.92
- Max cluster size: > 400
- Max time passed: 52 to 72 minutes

For the resolution service, the system's capability to suggest accurate resolutions was assessed based on the Standard Operating Procedures (SOPs) and knowledge base uploaded by the customer. Engineering team evaluations approved 95% of the system's resolutions as correct, underscoring its efficacy in providing relevant and accurate solutions to support engineers.

The Automatic Clustering and Resolution service significantly improved efficiencies, reducing manual effort and average resolution time by 30%, and cutting ticket backlog by 35%.

These gains underscore the system's capacity to streamline incident management processes, resulting in quicker and more effective resolution of IT issues, and demonstrating the substantial potential of the platform in enhancing operational performance across service management domains.

5 CONCLUSIONS

In conclusion, this research paper has demonstrated the potential of integrating Generative AI technologies into ticket management and monitoring systems. By leveraging these advanced technologies, we have successfully developed a system that allows organizations to easily transition from legacy ticket management practices. Our application-agnostic microservices offer a flexible and scalable solution that can be customized to meet the unique needs of various organizational environments. This modular system enables users to selectively adopt features and tailor the setup according to their specific operational needs. The observed efficiency gains further underscore the practical benefits of implementing such a system in a real-world IT environment.

The transition from legacy systems to this innovative platform, however, is not without its challenges. Resistance to change in organizations can be a significant barrier. Reducing the staff strength might be counterintuitive to the business model of many organizations. One size fits model all may have some unforeseen limitations. We might need to work closely with some organizations in the beginning/initially to make it robust enough to cover most business cases. Despite these hurdles, the benefits of adopting such a system—improved operational efficiency, reduced response times, and enhanced customer satisfaction—far outweigh the challenges.

As we continue to refine and expand our system, future research will focus on enhancing AI capabilities, developing incident predictive capabilities, optimizing the integration process, and expanding the system's adaptability to more diverse IT environments. By continuously advancing our technology and methodologies, we aim to set new standards in IT support operations, making them more proactive, predictive, and proficient. This research marks a

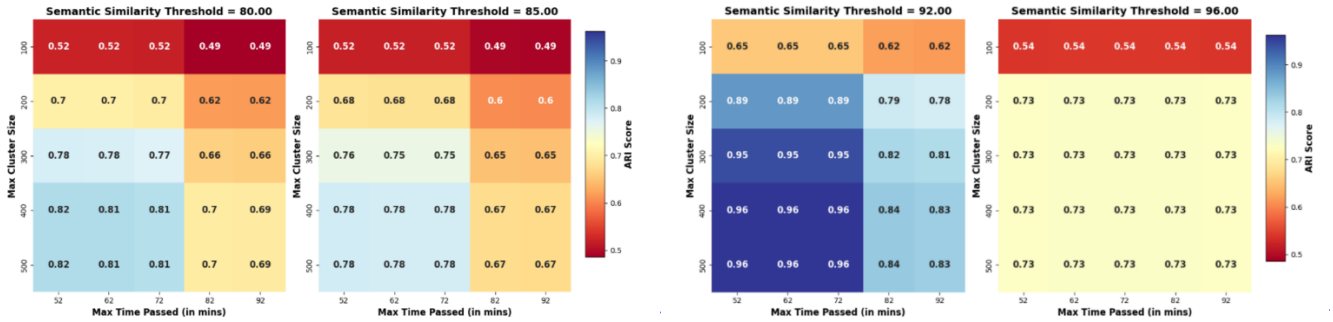

Figure 6: Hyperparameter tuning results.

significant step towards a future where IT support is more dynamic and driven by intelligent automation, paving the way for more resilient and responsive IT infrastructures.

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